Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Equations & Inequalities

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $3x - 7 = 2x + 5$. **[2]**

2. Solve $\dfrac{x+1}{2} = \dfrac{x-3}{3}$. **[3]**

3. Solve $\dfrac{2x-1}{3} + \dfrac{x+2}{4} = 1$. **[4]**

4. Solve $|x - 3| = 5$. **[3]**

5. Solve $|2x + 1| = 7$. **[3]**

6. Solve $|x + 2| < 5$. **[3]**

7. Solve $|3x - 1| \geq 4$. **[4]**

8. Solve $x^2 - 5x + 6 > 0$. **[4]**

9. Solve $x^2 + x - 6 \leq 0$. **[4]**

10. Solve $2x^2 - x - 3 < 0$. **[4]**

11. Solve the simultaneous equations $2x + y = 7$, $x - y = 2$. **[3]**

12. Solve the simultaneous equations $3x - 2y = 5$, $x + y = 4$. **[3]**

13. Solve $x^{3/2} = 8$. **[3]**

14. Solve $\sqrt{x+3} = x - 1$. **[4]**

15. Solve $\dfrac{1}{x} + \dfrac{1}{x+1} = 1$. **[4]**

Section B – Hard (Q16–25)

16. Solve $|2x - 1| = |x + 3|$. **[4]**

17. Solve $|x - 2| + |x + 1| = 5$. **[5]**

18. Solve $\dfrac{x+1}{x-2} > 3$. **[5]**

19. Solve $\dfrac{2x-1}{x+3} \leq 1$. **[5]**

20. Solve $x^2 - 3x - 4 \geq 0$. **[4]**

21. Solve the simultaneous equations $y = x^2 - 2x$ and $y = x + 2$. **[5]**

22. Solve $x^{1/3} + x^{-1/3} = 2$. **[5]**

23. Solve $\sqrt{2x-1}+\sqrt{x+4}=5$. **[6]**

24. Find the set of values of x for which $x^2+2x-8>0$ and $x>1$. **[4]**

25. Solve $\dfrac{x}{x+1}-\dfrac{2}{x-1}=1$. **[5]**

End of Worksheet • *Show all working clearly.*